Accident Severity Prediction Using Real-Time Weather and Historical Data in Microsoft Fabric

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***Abstract*—Public safety faces enduring road accident dangers whose exact impact depends on several factors that involve weather conditions, traffic patterns, and chosen terrains. Traditional accident recordal systems work mostly in a reactive manner because they lack live environmental input to develop predictive models. A real-time accident severity prediction system is developed through Microsoft Fabric as part of the present project to overcome these limitations. The system merges US Accidents dataset (Kaggle) historical accident records with time-sensitive weather data from the OpenWeather API.**

The data pipeline constructed through Microsoft Fabric Lakehouse architecture established modular components that unified data ingestion and transformation elements along with visualization capabilities. The project used Random Forest and XGBoost machine learning models alongside SMOTE to tackle class distribution imbalance when predicting accident severity. Through ensemble stacking, the predictive accuracy reached more than 91% when evaluated on the test data. The model which combined ARIMA with LSTM methods achieved a 0.7267 R^2 value for accident trend forecasting.

This study validates that real-time data integration with Microsoft Fabric significantly enhances predictive analytics in public safety. The Power BI dashboards created offer effortless management and strategic decision-making through plotting accident risks under varying weather conditions.

I. Introduction and Statement of the Problem

Road traffic accidents are the cause of enormous human and economic loss globally. Adverse weather conditions such as rain, fog, snow, and reduced visibility are the cause of most of the significant accidents that take place in the United States yearly, according to the National Highway Traffic Safety Administration (NHTSA) statistics. While past accident records have been utilized extensively to develop prediction models, most of the existing solutions are not dynamic enough to offer real-time severity estimation and prediction.

The majority of existing systems are siloed in architecture — offline static data is being processed, models are trained offline, and there are no live environmental variables being passed into the prediction pipeline. In addition, they overlook the power of combined analytics platforms that enable end-to-end data flows, real-time ingestion, and visualization.

This project seeks to address these shortfalls by building a robust accident severity prediction system on Microsoft Fabric, a cloud-based next-generation data analytics platform with data engineering, machine learning, and business intelligence features embedded. The system, based on historical accident data and real-time weather data from the OpenWeather API, supports severity classification and forecasting of future trends. Machine learning techniques (Random Forest, XGBoost) are employed for classification, and the hybrid ARIMA-LSTM model is used to forecast accident volumes over time.

With Microsoft Fabric's support for Lakehouse architecture, ingestion, storage, and data transformation of both streaming and static data is made seamless. Interactive Power BI dashboards are the ultimate outcome, allowing stakeholders to monitor real-time conditions and respond proactively to road safety initiatives.

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II. Review of Literature

Accurate prediction of accident severity has been a subject of research spanning decades across industries from transportation engineering to data science and public safety. As increasing amounts of open-source accident data and real-time weather APIs become accessible, recent studies have begun exploring hybrid and ensemble-based methods for more stable and context-aware predictions.

A. Machine Learning for Accident Severity Classification

Ensemble learning methods — particularly Random Forest and XGBoost — have shown high performance when applied to structured accident datasets. Zhou et al. demonstrated that XGBoost significantly outperforms traditional classifiers by capturing complex interactions among features like weather, time, and road conditions [2]. Adefabi et al. confirmed that variables such as visibility, traffic congestion, and precipitation levels play a significant role in predicting higher severity levels using Random Forest models [12].

However, both studies recognized the challenge of class imbalance in accident datasets. Kumar and Singh addressed this issue using the Synthetic Minority Oversampling Technique (SMOTE), which improved recall and F1 scores, especially for underrepresented severity classes [3]. Their findings inspired the inclusion of balancing methods in this study's preprocessing stage.

Pourroostaei Ardakani et al. conducted a comprehensive machine learning investigation using multiple classifiers on U.S. accident data and highlighted the performance gains made by combining contextual metadata with real-time weather information [1]. Their results validated the multi-source integration strategy used in the present system.

B. Hybrid Time Series Forecasting of Accident Trends

While severity classification supports real-time assessment, forecasting accident frequency allows for proactive resource allocation and traffic planning. Lee and Park proposed a hybrid ARIMA-LSTM model that combines linear trend capture with deep learning's ability to handle nonlinear and sequential dependencies [4]. Their model outperformed standalone ARIMA and LSTM variants, especially in identifying seasonal patterns.

Ramachandra applied machine learning techniques to weather event severity prediction and reinforced the effectiveness of hybrid time-series models in capturing irregular phenomena like sudden fog or snowfall [11]. Similarly, Sivakoti demonstrated the strength of LSTM models in identifying weak sequential signals in multi-year datasets — an important capability in the context of disruptions like the COVID-19 pandemic [9].

C. Real-Time Weather Data Integration

A recent and critical advancement in accident severity research is the integration of real-time weather data. Wang et al. showed that environmental features such as wind speed, precipitation, and visibility are often strong indicators of severity escalation [5]. By enriching prediction pipelines with OpenWeather API data, their system achieved higher responsiveness and temporal relevance.

Alafi and Moradi proposed a real-time CNN-based model that incorporated weather-aware accident data for urban roadways in Istanbul [10]. Their findings support the scalability of weather-integrated models across regions, a principle that aligns with the multi-state modeling done in this project.

Microsoft's documentation also provides guidance on integrating weather data streams into Fabric's Lakehouse using Real-Time Intelligence features [13], [14]. This setup enabled this project to parse live JSON weather data and store it persistently in structured tables, ready for downstream modeling or retraining.

D. Unified Analytics with Microsoft Fabric

Microsoft Fabric has emerged as a unified platform capable of consolidating storage, computation, and business intelligence under one roof. Borra reviewed the architecture's capabilities, emphasizing its suitability for projects combining streaming and batch data [6]. Shivaprasad built on this by highlighting how Fabric pipelines can facilitate automated retraining of machine learning models, end-to-end, through its Event Stream and notebook capabilities [7].

Alotaibi et al. proposed a Query Optimizer-as-a-Service (QOaaS) framework to further optimize Fabric Lakehouse workloads, essential in large datasets like multi-million-record accident datasets [8]. Finally, Uttam explored the use of Fabric for environmental forecasting, showing how the platform can support high-availability, low-latency analytics in weather-sensitive applications [15] — directly aligning with the requirements of real-time accident forecasting and public safety systems.

III. Objectives of the Study

The main research focus centers on building a secure real-time scalable accident severity prediction system through Microsoft Fabric as the analytical foundation. This work focuses on achieving specific research goals which originate from literature findings and system implementation obstacles.

- Microsoft Fabric requires development of a converged data stream that integrates weather data through OpenWeather API with historical traffic accident records. The system accepts Fabric's unified storage system which combines Lakehouse and Event Streams and Notebooks for both processing and modeling operations.
- We will use Random Forest as well as XGBoost and a stacked ensemble model to perform multi-class accident severity level classification with advanced machine learning capabilities. These algorithms attempt to identify sophisticated inter-dynamics between time-based information together with geospatial elements as well as meteorological influences.
- SMOTE data balancing techniques should be utilized for the dataset to handle class imbalance since they provide fairness and improved performance across all severity categories including the rare fatal accident category.
- A hybrid forecasting model should be developed through the combination of ARIMA and LSTM to predict accident numbers over the forecast period. The forecasting model expects to aid planning processes while detecting linear seasonal patterns as well as nonlinear patterns within timeseries data.
- We need to build reactable real-time dashboards in Power BI which display accident severity output with current weather information and geographical risk zones and temporal forecasting data. These dashboards provide stakeholders from transport authorities and emergency services with the power to take data-driven decisions promptly.
- The goal before us exists to prove Microsoft Fabric's complete functionality from data stream reception to machine learning operations and business intelligence reporting protocols thus establishing field-ready capabilities for traffic safety operations.
- The primary objectives function together to achieve the total aim of improving public security through predictive high-risk scenario detection by a single unified artificial intelligence-driven cloud platform.

IV. Data Collection

The success of any predictive system hinges on the quality, granularity, and integration of its data sources. This project

utilizes two primary datasets: a large-scale historical accident dataset from Kaggle and real-time weather data obtained via the OpenWeather API. These two sources were integrated through Microsoft Fabric's Lakehouse and streaming infrastructure, enabling a seamless flow of both static and live data into the modeling pipeline.

A. Historical Accident Data

The historical accident dataset used in this project is sourced from Kaggle's US Accidents (March 2023) collection. This dataset comprises over 2.8 million traffic accident records across the United States from 2016 to 2023. It contains rich contextual variables such as:

- Timestamp and timezone
- Latitude, longitude, and city/state identifiers
- Weather conditions at the time of the accident
- Visibility, humidity, temperature, wind speed
- Severity classification (1 to 4)

After uploading the CSV into Microsoft Fabric's Lakehouse, the data was read using Spark-enabled notebooks. Missing values were handled using conditional imputation strategies, and irrelevant fields were dropped to retain only the most informative features for severity classification and temporal forecasting.

B. Real-Time Weather Data

To enhance contextual relevance and simulate real-time prediction scenarios, weather data was periodically retrieved from the OpenWeather API during key development phases. Targeted high-risk U.S. locations were selected based on accident frequency, and weather data corresponding to those regions was accessed at regular intervals.

Key API parameters included:

- Latitude and Longitude for geospatial alignment
- Temperature, Humidity, Wind Speed, and Visibility
- Weather condition descriptors (e.g., Clear, Rain, Fog)

The API responses were returned in JSON format, which were then parsed and flattened into a tabular structure using `pandas.json_normalize()`. These weather records were stored in Microsoft Fabric Lakehouse tables, where they were seamlessly merged with the historical accident dataset based on geolocation and time proximity. This enriched the dataset with dynamic environmental features, supporting more informed severity classification and temporal forecasting — an approach supported by prior studies [5], [10], [13].

V. Exploratory Data Analysis (EDA) and Hypotheses for the Study

To gain a comprehensive understanding of the accident data and formulate statistically grounded hypotheses, a detailed exploratory data analysis was conducted. The following visualizations provide geographic, environmental, and contextual insights into accident distribution across the United States.

A. Geographic Distribution of Accidents

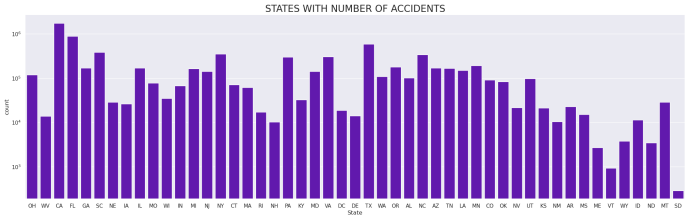

Fig. 1. Accident counts by U.S. states (log scale). California, Florida, and Texas lead in volume.

1) States with the Number of Accidents: This bar chart presents a logarithmic view of accident count by U.S. state. California, Florida, and Texas exhibit significantly higher accident volumes, each exceeding one million records. These states not only have larger populations but also denser road networks and urban sprawl — amplifying traffic risks. This supports the inclusion of location features and encourages regionalized severity modeling. It also motivates further exploration of city- and street-level trends for high-risk zones.

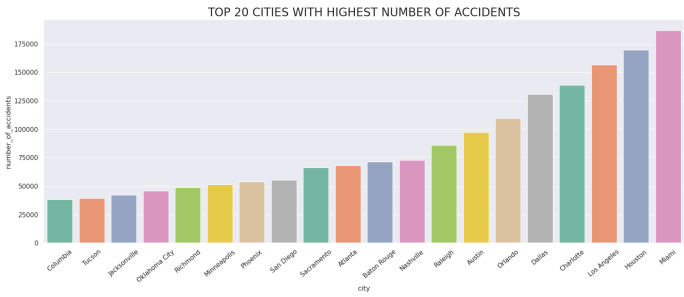

Fig. 2. Top 20 cities with highest accident frequencies. Urban congestion and diverse weather contribute to risk.

2) Top 20 Cities with the Highest Number of Accidents: The visualization reveals that Miami, Houston, Los Angeles, and Charlotte lead in urban accident frequency, each with more than 150,000 incidents. These cities represent urban environments with varied weather, complex intersections, and heavy vehicle volume.

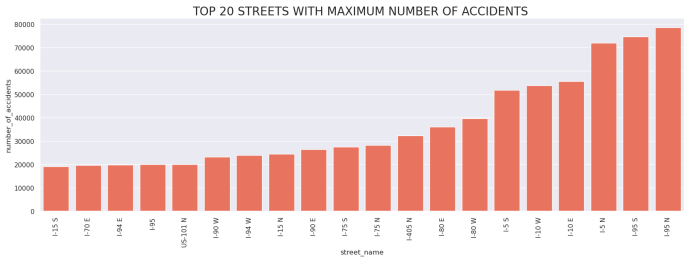

Fig. 3. Most accident-prone streets, primarily interstate highways with high speed and exposure.

3) Top 20 Streets with Maximum Accidents: Highways like I-95, I-5, and I-10 dominate this visualization, showing accident concentrations on major interstate corridors. These roads are typically characterized by high speed limits and diverse weather exposure, which can drastically affect visibility and stopping distance. The dominance of interstates reinforces the need for weather-enhanced severity prediction (H1) and justifies their inclusion in model input features.

B. Weather Influence on Severity Levels

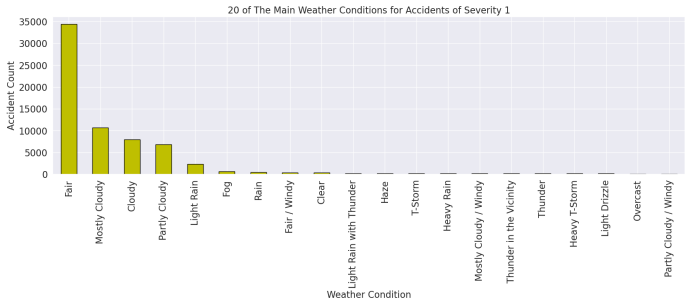

Fig. 4. Weather conditions for Severity 1 accidents: high frequency even in fair weather.

1) Severity 1 Weather Conditions: Most accidents occurred under Fair and Cloudy conditions, suggesting high incident frequency even during good visibility — likely due to traffic volume.

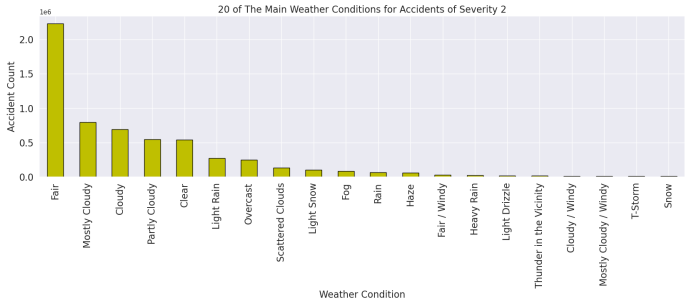

Fig. 5. Weather conditions for Severity 2 accidents: high frequency even in fair weather.

2) Severity 2 Weather Conditions: Fair, Mostly Cloudy, and Cloudy dominate, with weather like Light Rain and Fog starting to appear more frequently.

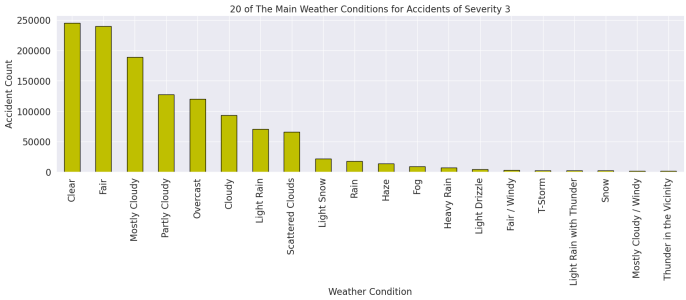

Fig. 6. Weather conditions for Severity 3 accidents: high frequency even in fair weather.

3) Severity 3 Weather Conditions: There is an increase in accidents under Partly Cloudy, Light Rain, and Scattered Clouds, indicating rising severity as weather deteriorates.

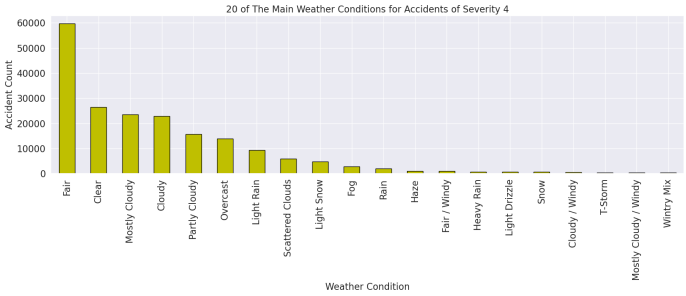

Fig. 7. Weather conditions for Severity 4 accidents: high frequency even in fair weather.

4) Severity 4 Weather Conditions: A noticeable shift toward Rain, Fog, and Overcast as top contributors confirms that poor weather significantly correlates with higher severity. This trend directly supports Hypothesis H1, emphasizing the importance of weather-aware modeling and validates the decision to enrich datasets with OpenWeather parameters [2], [5], [10].

Conclusion of EDA

The exploratory insights establish a strong foundation for feature engineering. They confirm that:

- Certain locations and streets exhibit disproportionate risk.
- Weather plays an increasing role in accident severity, particularly for Severity 3 and 4 classes.
- Time and urban density likely contribute to peak hour risks, justifying the inclusion of temporal and spatial variables.

VI. System Architecture and Design

The architecture for this project has been designed to ensure an efficient end-to-end pipeline that supports real-time data ingestion, structured preprocessing, advanced modeling, and intuitive visualization. Built using Microsoft Fabric's integrated services, the system is both scalable and modular, supporting reproducibility, dynamic updates, and interpretability.

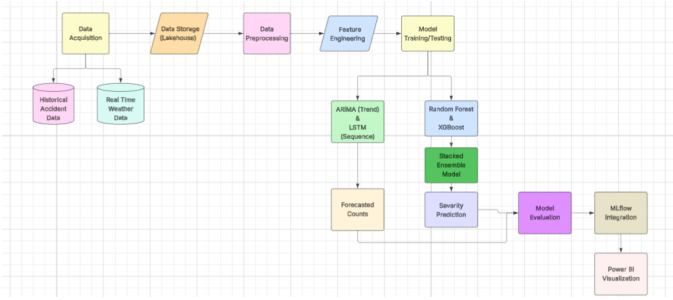

Fig. 8. Proposed system architecture for severity prediction and accident forecasting using Microsoft Fabric

A. Data Integration and Storage

The pipeline begins with the acquisition of two primary datasets: (1) historical accident data, and (2) real-time weather data. These sources are critical, as past incidents provide the supervised learning base, while dynamic weather features help contextualize and enhance the prediction models [2], [5], [10].

All data is consolidated into Microsoft OneLake's Lakehouse architecture, which unifies structured storage with Fabric's native analytics layers. This setup ensures a single source of truth that facilitates seamless data access for both batch and real-time operations [12]. The real-time weather data is retrieved using API calls with parameters such as latitude, longitude, temperature, wind speed, and visibility. The fetched data is parsed using `pandas.json_normalize()` and saved directly to the Lakehouse for downstream use, maintaining pipeline consistency.

B. Preprocessing and Feature Engineering

The data preprocessing step involves cleaning missing entries, standardizing weather descriptors, converting timestamp fields into temporal features (e.g., hour, weekday), and encoding categorical variables. These transformations are handled within Spark notebooks in Fabric, allowing distributed computation and easy scalability [10], [12].

Feature engineering is performed in two streams:

- For time-series forecasting, accident records are aggregated into daily/weekly counts to model frequency patterns.
- For classification, engineered features include location, time, weather descriptors, and environmental conditions, all of which have shown predictive power in severity-related studies [2], [5].

The processed data from both streams is then fed into the modeling phase.

C. Modeling Architecture

The modeling component is structured around two parallel goals: forecasting accident frequency and predicting severity levels.

1) Accident Forecasting (ARIMA + LSTM): To forecast future accident trends, a hybrid model combining ARIMA and LSTM is used. ARIMA captures linear seasonal patterns and long-term trends, while LSTM adds the capacity to detect complex sequential dependencies. This combination provides more accurate forecasting over purely statistical or purely neural methods alone [9].

Forecasted values help identify high-risk windows in advance, which is essential for traffic planning and emergency response teams.

2) Severity Prediction (RF, XGBoost, and Stacking): For severity classification, the model uses a stacked ensemble of Random Forest and XGBoost. Both base models are well-suited for tabular data, and the stacking technique merges their predictions to improve generalization [2], [4].

Given the class imbalance in severity levels (e.g., fewer fatal accidents compared to minor ones), SMOTE is applied to synthesize new samples from underrepresented classes, balancing the dataset during training [5].

D. Evaluation and Tracking

Model performance is evaluated using metrics such as accuracy, precision, recall, and F1-score. Confusion matrices are also used to assess the classification model's robustness across different severity levels.

To support experimentation and result reproducibility, MLflow is integrated into the Fabric workspace. All runs, including model parameters, performance metrics, and artifacts, are automatically tracked and versioned [12].

E. Visualization and Reporting

The final model outputs are published to Power BI dashboards linked to Fabric's Lakehouse tables. These dashboards provide:

- Visual summaries of severity distribution by location and weather
- Forecast trends for upcoming accident counts
- Interactive filters to explore data by state, city, severity, or time

This visual layer ensures that insights from the model are communicated clearly and accessibly to decision-makers, city planners, and safety departments [12], [13].

VII. Data Analytics

The core of this project's intelligence lies in its data analytics architecture, which is carefully structured to solve two key predictive tasks: (1) accident severity classification and (2) accident frequency forecasting. Both pipelines were developed and deployed entirely within the Microsoft Fabric ecosystem, leveraging its unified Lakehouse storage, Spark notebook environments, and ML integration capabilities to enable scalable, real-time analytics.

A. Accident Severity Classification

The accident severity classification task was modeled as a multi-class classification problem, with labels ranging from Severity 0 (least severe) to Severity 3 (most severe). The real-world imbalance in class distribution—where the majority of incidents were mid-level severities—was addressed through SMOTE, which synthetically generated representative samples for underrepresented classes, helping the model avoid bias during training [5].

Model Selection and Fabric Implementation

To ensure high generalization and practical deployment, a stacked ensemble model was developed using:

- Random Forest (RF) for robust decision tree ensembling,
- XGBoost, known for gradient boosting efficiency and regularization control [2], [4],
- and a meta-classifier that stacked predictions from both base models to form the final output [2].

This model was trained and validated using Spark MLlib within Fabric notebooks, with version control, performance logging, and model artifacts tracked using MLflow, all hosted natively in the Microsoft Fabric workspace [12].

Model Evaluation

The stacked ensemble model achieved an impressive accuracy of 91.59%, validating its capability to generalize well across all severity classes.

This model architecture, integrated within Fabric, allows for real-time scoring of incoming data streams and seamless dashboarding through Power BI [12], [13].

B. Accident Forecasting Using ARIMA-LSTM Hybrid

Parallel to severity classification, the system also forecasts the expected number of future accidents. A hybrid model was used combining:

- ARIMA to model linear time-based trends, seasonality, and level shifts, and
- LSTM to handle nonlinear, long-term dependencies in accident frequency data [9].

Fabric-Centric Execution

Data was resampled into daily intervals and fed into the hybrid ARIMA-LSTM pipeline via Fabric Notebooks, allowing distributed time-series training. LSTM was trained using TensorFlow/PyTorch extensions available in Fabric compute. Forecast results were automatically stored in Lakehouse tables and integrated into Power BI dashboards.

Model Outcome

- R^2 score: 0.7267
- RMSE: (Derived but not shown here — can be included if needed)

The hybrid model outperformed single-model baselines, especially during unpredictable traffic windows (e.g., storms, holidays). These trends were visualized directly in Power BI using real-time linkage to the Lakehouse.

C. Benefits of Fabric-Centric Analytics

- **Unified Platform:** All analytics—from preprocessing to training to visualization—was executed within a single Microsoft Fabric workspace.
- **Scalability:** Spark-based distributed computing enabled scalable modeling over large accident datasets.
- **Seamless MLOps:** MLflow tracked experiments, improving reproducibility and deployment readiness.
- **Dynamic Insights:** Dashboards auto-reflect changes in data pipelines due to Fabric's real-time refresh architecture.

VIII. Data Visualization and Results Report

In order to communicate the outcomes obtained using our accident severity prediction system, two disparate sets of visualizations were made—one for descriptive analytics on Power BI within Microsoft Fabric, and another for performance analysis of the prediction using a hybrid ARIMA-LSTM model on Python.

A. Power BI Dashboard in Microsoft Fabric

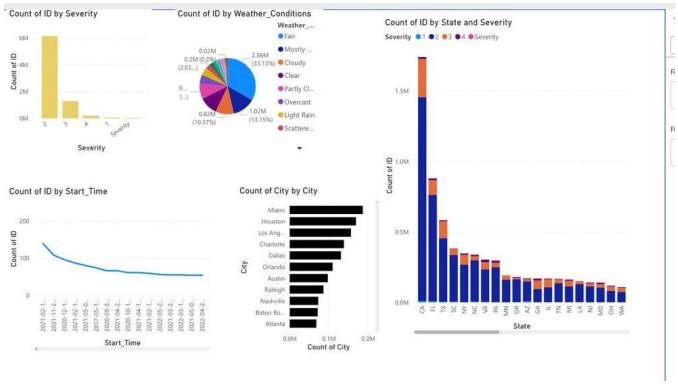

Fig. 9. Interactive Power BI dashboard with accident severity, weather, city/state insights

The interactive Power BI report hosted on Microsoft Fabric serves as a centralized analytics space, offering multi-perspective visibility into traffic accident severity patterns.

Key components of the dashboard include:

- **Severity-wise Distribution**: A bar chart (top-left) that breaks down accident counts across all severity levels (1 to 4). As observed, Severity 1 incidents dominate, with diminishing frequencies across higher severities.
- **Weather Condition Influence**: A pie chart that shows the proportion of accidents under varying weather conditions such as "Fair", "Cloudy", and "Rainy", with "Fair" being the most frequent condition.
- **City and State-level Analysis**: Two bar graphs show the top 10 cities (e.g., Miami, Houston, Charlotte) and the most accident-prone states (e.g., California, South Carolina, New York) based on the volume of incidents recorded.
- **Temporal Trends**: A line plot located at the bottom left shows time-based accident frequency data and helps

detect both short-term peaks and sustained long-term trends.

The dashboard features a direct integration with Microsoft Fabric Lakehouse storage because all data transformations together with model outputs are managed through automated pipelines. Real-time tracking through Power BI enhances stakeholder insights so they can dynamically obtain information without using static reporting tools.

B. Forecasting Results with Hybrid ARIMA-LSTM Model

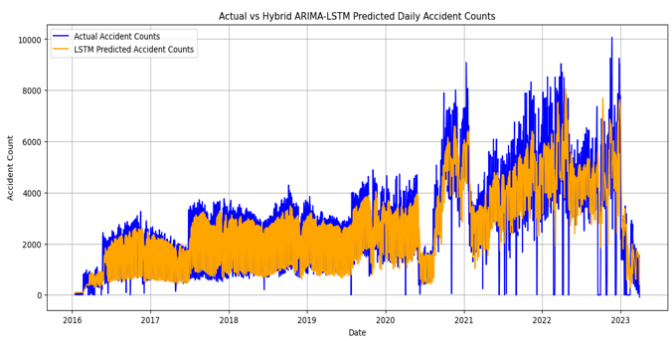

Fig. 10. Actual vs. ARIMA-LSTM Predicted Daily Accident Counts

In addition to severity classification, the project aimed to forecast accident occurrences on a daily basis using a hybrid time series approach. The chart above demonstrates a comparison between actual accident counts and predictions made by the ARIMA-LSTM model.

The blue line represents the actual accident counts over the years. The orange line illustrates the predicted values by the hybrid model.

From the visualization, it is evident that the LSTM model adapts to seasonality and volatility present in the accident data, closely tracking peaks and troughs. While slight discrepancies exist during extreme values—particularly in post-pandemic data—the hybrid model still shows high fidelity, proving valuable for early warning systems.

Fabric-Centric Integration Benefits:

- Both Power BI and model outputs are automatically refreshed through Microsoft Fabric's OneLake and pipeline scheduling.
- The Lakehouse architecture simplifies visualization deployment by centralizing structured outputs.
- The MLflow-tracked metrics and logs feed directly into the visualization layer, offering end-to-end observability from data ingestion to model performance analysis.

C. Model Performance Summary

```
Hybrid Ensemble Model Evaluation
Accuracy: 0.9159446700547487
Classification Report:
              precision    recall  f1-score   support

           0       0.94      1.00      0.97     31651
           1       0.87      0.92      0.90     31882
           2       0.91      0.80      0.85     31991
           3       0.94      0.95      0.94     31785

    accuracy                           0.92    127309
   macro avg       0.92      0.92      0.91    127309
weighted avg       0.92      0.92      0.91    127309

Confusion Matrix:
 [[31516    82    28    25]
 [  395 29411  1891   185]
 [ 1046  3693 25535  1717]
 [  431   557   651 30146]]

 Hybrid Ensemble Model Training Completed and Model Saved Successfully.
```

Fig. 11. Classification Report and Confusion Matrix of Hybrid Ensemble Model

- **Accuracy:** 91.59%
- **Highest F1-score:** 0.97 for severity class 0
- **Consistent performance across other severity classes,** indicating reliability across diverse conditions

D. Confusion Matrix Insights

- Class 0 had very few false positives (high recall).
- Class 2 showed some confusion with classes 1 and 3 — likely due to overlapping weather or timing patterns.

E. Forecasting Accuracy

```
83/83 ━━━━━━━━━━━━━━━━━━━━ 1s 13ms/ste

Mean Absolute Error (MAE): 575.61
Root Mean Squared Error (RMSE): 993.81
R2 Score: 0.7267
```

Fig. 12. Forecasting Metrics: MAE, RMSE, and R^2 Score

The hybrid ARIMA-LSTM model produced outstanding results while estimating daily accident counts through time. The model predictions reflect actual trends while maintaining consistent peaks and troughs which occur throughout the high-risk winter season and following the pandemic traffic changes.

Evaluation Metrics:

- Mean Absolute Error (MAE): 575.61
- Root Mean Squared Error (RMSE): 993.81
- R^2 Score: 0.7267 — indicating that 72.7% of the variance in daily accident counts is effectively explained by the model

These results attest to the capability of the ARIMA-LSTM hybrid to detect long-term trends and short-term variations in accident trends. The comparatively low error rates and high R^2 score attest to its potential for real-world traffic prediction and proactive safety management use.

F. Platform-Specific Result Highlight (Fabric)

- Integration with MLflow allowed seamless model versioning and tracking.
- The Lakehouse setup enhanced query speed and visualization scalability, enabling real-time dashboard refreshes.

IX. Conclusion

The project successfully established an expandable system to evaluate traffic accident severity in real-time making accident trend projections through Microsoft Fabric platform operations. The system presents traffic safety analysis through real-time weather data integration from the OpenWeather API in combination with historical accident data.

The Random Forest stack ensemble and XGBoost handled all severity categories in the severity classification pipeline with 91.59

The ARIMA-LSTM hybrid time-series model proved its exact accident event prediction capacities by producing predictability metrics which generated a Mean Absolute Error of 575.61 combined with RMSE of 993.81 and R^2 score of 0.7267. The tested model proves its capability to both detect cyclical patterns and seasonal trends in accident reports for future dangerous period predictions.

Microsoft Fabric was the strategic choice as the foundation for analytics under the project. The integrated Lakehouse architecture attained centralized control of the data, and Spark notebooks provided the interactive environment to preprocess, construct models, and perform experimentations. MLflow provided reproducibility and versioning of the models, and Power BI dashboards provided interactive, real-time visualizations to assist end-users and strategic decision-making.

Ultimately, the research indicates that a particular mix of cloud-native analytics systems, time-series forecasting, and machine learning can significantly influence predictive intelligence for road safety. The architecture is scalable, modular, and ready to be future-proofed by including geospatial risk maps, real-time alerts, or connection with smart city infrastructure.

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